Two further field-extension structural tables

Private recovery component

Theorem 1. *All 47 displayed structural descriptions in the current companion table for* $\mathrm{L}_2(16) : 2$*, and all 30 in the table for* $\mathrm{P\Gamma L}_2(32)$*, hold for the actual subgroup representatives specified by the accepted numerical row bijections. The top rows receive actual split-extension certificates. An extension description asserts the indicated decomposition, not unique determination of an isomorphism type by its factor names.*

Exact source and model interfaces. These two tables are attributed to Thomas Connor and Dimitri Leemans. All four native pages were inspected, including the empty continued table on the second $q = 32$ page. Current bytes are retained, without asserting identity with all historical companion editions. The complete numerical inventories and source-row bijections are acquired results. No lattice-property premise is inferred from the headings.

The native ledger distinguishes cyclic order 33 at $q = 32$, row 15, from the elementary abelian powers 2^e, $e = 2, 3, 4, 5$. For $q = 16$, row 14 is $(2^3 \cdot 2) : 2$, row 23 is $(2 \times 4) : 2$, and row 9 is $D_{34} \cdot 2$. For $q = 32$, row 5 is $(11 : 5) : S_3$ and row 8 is $(2^4 : 5) \times 2$. All products associate from the left, as specified in Section 3 of the acquired 2015 naming article. A dot denotes nonsplitting, a cross direct product, and a colon a split extension. The source targets do not select between alternative actual models of the same description.

The $q = 16$ representatives are extracted with the exact acquired strict literal-state parser from the completed normal-restriction state. Its named-model transport identifies the target with the projective $\mathrm{SL}_2(16)$ action extended by the square of the field Frobenius; the full order is 8160. The already accepted transport and its full verification are acquired here. For $q = 32$, every subgroup and ambient group is parsed from literal embedded permutation cycles in the saved complete inventory. The originally executed ambient model was the permutation image of $\mathrm{Aut}(\mathrm{PSL}(2, 32))$. The named-constructor and classical semilinear identification remain explicit inherited premises. No proper subgroup is reconstructed from a display name, and group order alone is never used as its identification.

Shared certificates. The accepted PSL_2, PGL_2 and preceding semilinear components supply 41 required concrete models and descriptions. Two previously accepted PSL_2 ambient actions supply the simple factors $\mathrm{L}_2(16)$ and $\mathrm{L}_2(32)$. The present evidence adds 18 models; every factor reference points to an acquired or earlier verified model. The acquired MC.3 generator-edge criterion checks each actual transport: a bijection fixes identity and respects right multiplication by every ordered generator, hence is a homomorphism by induction on words. No full quadratic multiplication table is required.

Actual decomposition tests. All criteria are those already proved in the acquired components. The literal verifier checks the order and complete closure of each group. For a split node $H = N : K$, it checks the two factor descriptions by actual generator-edge bijections, normality of N, $N \cap K = 1$ and $|N||K| = |H|$. A noncommuting generator pair checks nontrivial action. Composite complements, including the S_3 in $(11 : 5) : S_3$, undergo these same tests. Prime-five complements in the field-extension top group and its solvable subgroups are also tested literally.

For a direct node, all factor generators commute; together with the intersection and product-size checks, this proves the displayed direct product. There are seven new direct-product models. Each dotted step has index two and is checked by the complete outside-involution obstruction: a complement exists exactly when an outside element has order two. The new model $2^3 \cdot 2$ has eight outside elements, all with nonidentity square. Other dotted descriptions transport already accepted models. No failure of a search routine is used as a nonsplitting proof.

Finite evidence and checks. The single bounded witness pass returned all 77 row certificates in 2.37 seconds, without an ambient subgroup census. Normal-subgroup, complement and isomorphism routines only suggest data. The independent literal verifier checks all 77 source joins, all 18 new models, 112 transports and 679,553 right-generator edges. The two top rows have full certificates, and every source row is tied to its exact saved embedded generators before a structural assertion is checked.

The damaged controls alter identity images, a bijective nonhomomorphism, element and row coverage, source row maps, actual generators, the cyclic 33 and power 2^5 distinction, a centered dot, product association, quotient index, acquired models, complement generators, dependency order and a direct factor. Each fails an explicit guard. Independent all-element controls distinguish $C_3 \times C_2$ from the noncommuting factors in S_3; prime-index controls distinguish C_{p^2} and C_p^2 for $p = 2, 3$. Ordinary and optimized Python must agree. The complete literal state parser is acquired unchanged, including its rejection of nonliteral constructors and invalid embedded cycles.

Scope and attribution. The finite result concerns 77 exact current structural descriptions, conditional on accepted complete subgroup inventories, numerical source bijections and named-model interfaces. It does not assert unique extension naming, a new subgroup classification, completeness of the whole atlas, or identity of unseen editions. Standard product and transport arguments are acquired; the new evidence consists of the fixed native ledger, decomposition witnesses and literal joins. Root authorized continued registry reuse; the component author prepared these actual certificates and checks. No original-source, question, formalization or measurement count changes here.